

ANAND MEENA

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EDUCATION

University of Illinois Chicago

Master of Science in Computer Science (GPA: 4.0)

Coursework: Cloud Computing, Natural Language Processing, Information Retrieval, Data Science, Data & Algorithmic Fairness

Netaji Subhas University of Technology

Bachelor of Technology in Computer Science and Engineering

Chicago, IL

August 2024 – May 2026

New Delhi, IN

August 2019 – May 2023

EXPERIENCE

University of Illinois Chicago

Software Engineer – SAMPL (Searles Applied and Materials Physics Laboratory)

Chicago, IL

December 2025 - Present

- Leading development of ML-based quantum state tomography (QST) systems on NVIDIA DGX infrastructure, collaborating with faculty on research targeting publication in quantum computing conferences and peer-reviewed journals.
- Designed and executed a large-scale empirical study benchmarking classical neural network (MLP, GRU) performance on quantum state reconstruction across 1–10 qubits; identified fidelity breakdown thresholds (< 0.7 for mixed states at 5–6 qubits) using ~140 GB of GPU-generated Pauli measurement datasets.
- Engineered a hardware-accelerated ML inference pipeline using HLS4ML to deploy GPU-trained neural networks on a Xilinx ZCU216 RFSoc FPGA, integrating real-time quantum measurement acquisition with models trained on NVIDIA DGX infrastructure.

Oplexa AI Consulting & Market Research

Software Engineer

Santa Clara, CA

June 2025 – December 2025

- Developed MVPs and early prototypes for cloud infrastructure optimization platforms for investor demonstrations, architecting foundational features, and building proof-of-concept solutions targeting Fortune 1000 enterprise deployments.
- Architected a scalable digital twin visualization system targeting \$20B+ in annual infrastructure waste, integrating Three.js-based 3D scene rendering with 200+ configurable parameters to simulate real-world datacenter layout and resource behavior
- Built a multi-cloud IaaS platform for LLM inferencing optimization, designed to deliver 30-60% cost savings across 3,000+ Fortune 1000 enterprises with \$1.4M+ GPU budgets
- Engineered SLA-aware workload orchestration with real-time NVIDIA and Kubernetes telemetry integration, automating deployment processes and reducing provisioning time from weeks to 15 minutes
- Developed unified control plane with REST API interfaces supporting federated compute management across 10+ cloud providers and enterprise data centers
- Delivered technical demonstrations to investors and stakeholders, contributing to early-stage fundraising and product validation efforts

Hans Informatics Pvt Ltd

Software Development Intern

New Delhi, IN

August 2022 - October 2022

- Developed and deployed a real-time Android application for proctored examinations using ML Kit, TFLite, and FaceNet to authenticate users and enforce location-based access
- Attained 95% face-matching accuracy by optimizing preprocessing pipelines and applying bitmap augmentation across heterogeneous input conditions (varying lighting, angles, and device cameras).
- Integrated GPS-based localization (~10m precision) with facial recognition to ensure spatial identity compliance for test-takers
- Designed an anomaly detection module to flag behavioral discrepancies in real-time, minimizing manual supervision requirements

Solarman Technology Project Pvt Ltd,

Software Development Intern

New Delhi, IN

April 2022 – May 2022

- Architected a modular Django platform with PostgreSQL for ~1,200 records/month across 20+ clients; added REST API endpoints and an admin dashboard, reducing client onboarding time.
- Improved UI performance using AJAX, CSS, and responsive layout optimizations, reducing page load latency by ~30%.
- Integrated Amazon SES with token auth to automate password resets and recovery, reducing access tickets by 8%.
- Deployed production infra on AWS Elastic Beanstalk and RDS with autoscaling and CloudWatch alerts; maintaining 99.7% uptime for 30-50 daily active users.

PUBLICATIONS & PROJECTS

Cross-Language Question Answering System Using Hugging-Face Transformers

Published in Springer CCIS 2234, ICICBDA 2024 Conference Proceedings

January 2023 – June 2024

- First author on peer-reviewed conference paper published in Springer Communications in Computer and Information Science (CCIS 2234), presenting novel cross-language QA framework addressing gaps in transformer-based multilingual systems
- Developed a novel cross-language QA framework addressing critical gaps in transformer-based multilingual systems, demonstrating improved cross-lingual transfer capabilities for low-resource language pairs.

- Achieved 92% QA accuracy with translation fidelity scores of 35.9-41.3 BLEU across language pairs, demonstrating superior performance in cross-lingual information retrieval tasks
- Published findings demonstrate practical deployment of transformer models for both translation and question-answering, contributing to advancement of multilingual NLP accessibility

Distributed AI Pipeline for LLM Preprocessing – CS441

August 2024 - December 2024

- Orchestrated preprocessing, tokenization, and streaming transformations across HDFS and MapReduce stages for scalable LLM training
- Designed a Hadoop-based NLP pipeline for preprocessing 40GB of LLM training data using Scala and shell automation, achieving 3x faster processing compared to single-node implementations through distributed computing architecture
- Built gRPC-based connectors between Akka HTTP on EC2 and AWS Lambda to enable fast, modular inference calls with sub-100ms latency for real-time applications.
- Applied Byte Pair Encoding and context-preserving sliding window tokenization (stride: 128 tokens) to support multi-segment input reconstruction.
- Integrated LLaMA3 into a serverless Bedrock API stack for multi-turn question answering, deployed comprehensive monitoring via CloudWatch with error tracing and latency alerts reducing debugging time by 60%.

SKILLS

Languages: Python, JavaScript, HTML, CSS, C/C++, Scala, SQL (PostgreSQL), PHP, R

Cloud & Infrastructure: AWS (Lambda, EC2, RDS, Bedrock, CloudWatch), Azure, GCP, Kubernetes, Docker

Machine Learning & AI: TensorFlow, Hugging Face Transformers, LLaMA3, DistilBERT, TFLite, ML Kit, FaceNet, HLS4ML

Development & Tools: Git, VS Code, PyCharm, IntelliJ, Flask, Django, REST API, gRPC, Three.js, AJAX

Data & Systems: Hadoop, HDFS, MapReduce, PostgreSQL, SciPy, Matplotlib, OpenCV, Ubuntu, NVIDIA GPU technologies, Xilinx FPGAs